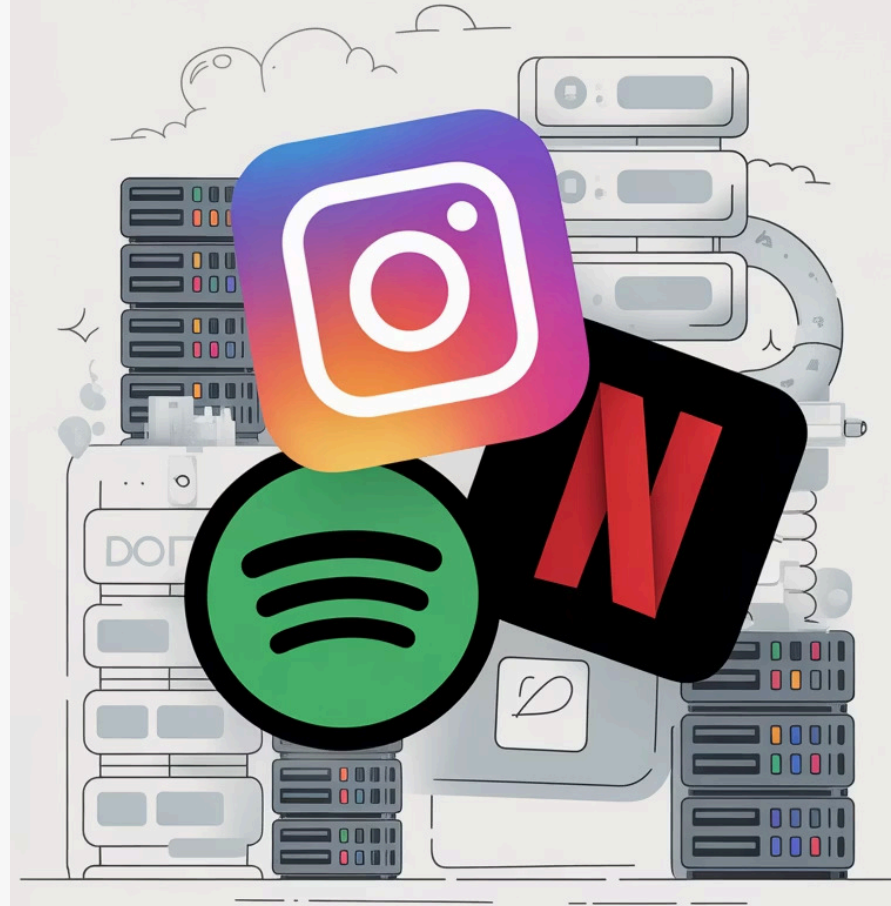


Introduction to Modern Apps

What powers apps like Instagram, Netflix, or Spotify?

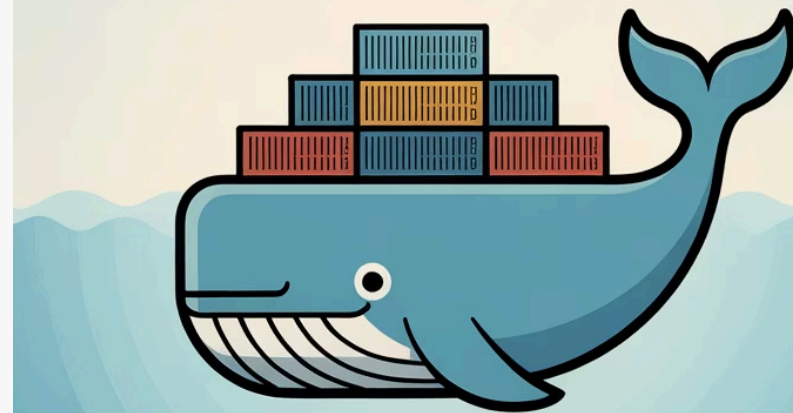
- They run on many servers 
- Need to scale fast 
- Must stay online 24/7 

by Naresh Chaurasia



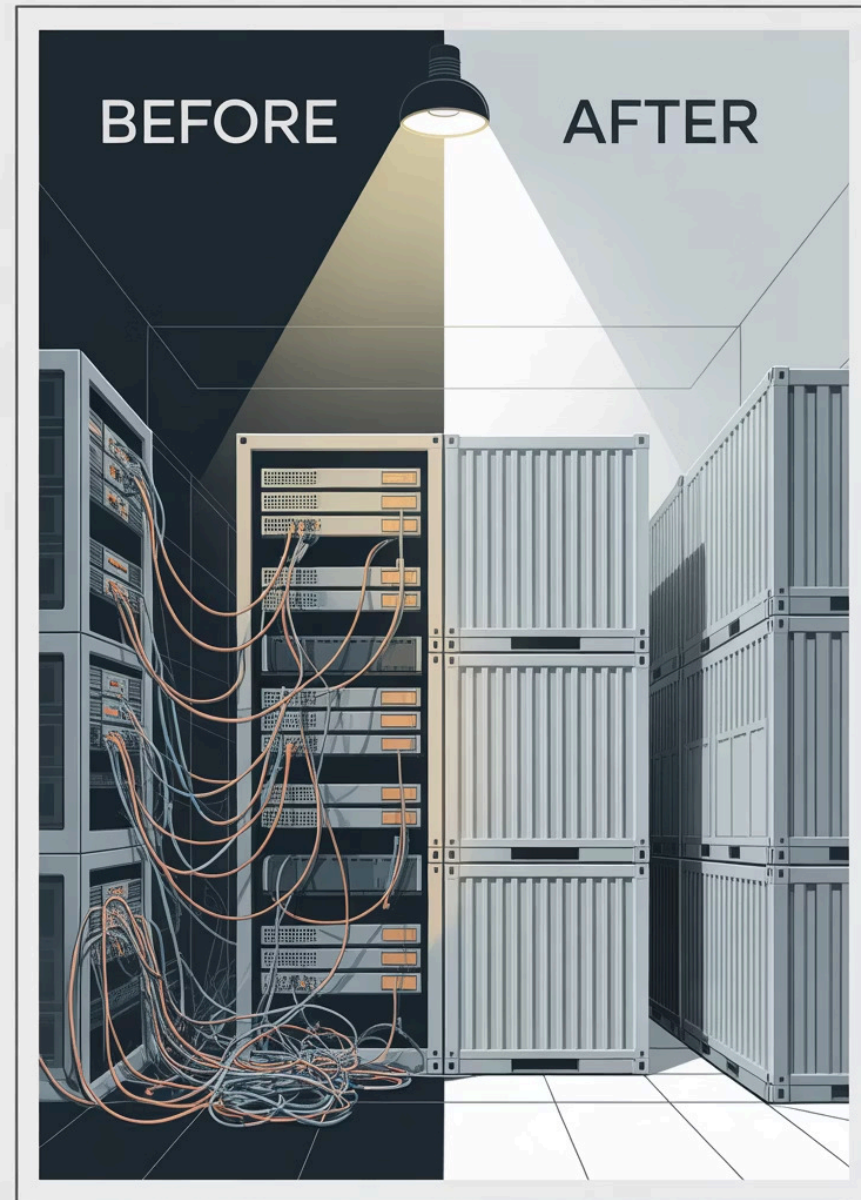
Meet Docker – The Container Hero

- Docker = Package your app + dependencies 
- Runs anywhere: laptop, server, or cloud 
- Lightweight, faster than Virtual Machines 



Kubernetes: Why Containers Matter

- Consistency: "Works on my laptop" ✓
- Easy sharing across teams 🤝
- Faster deployments ▶▶





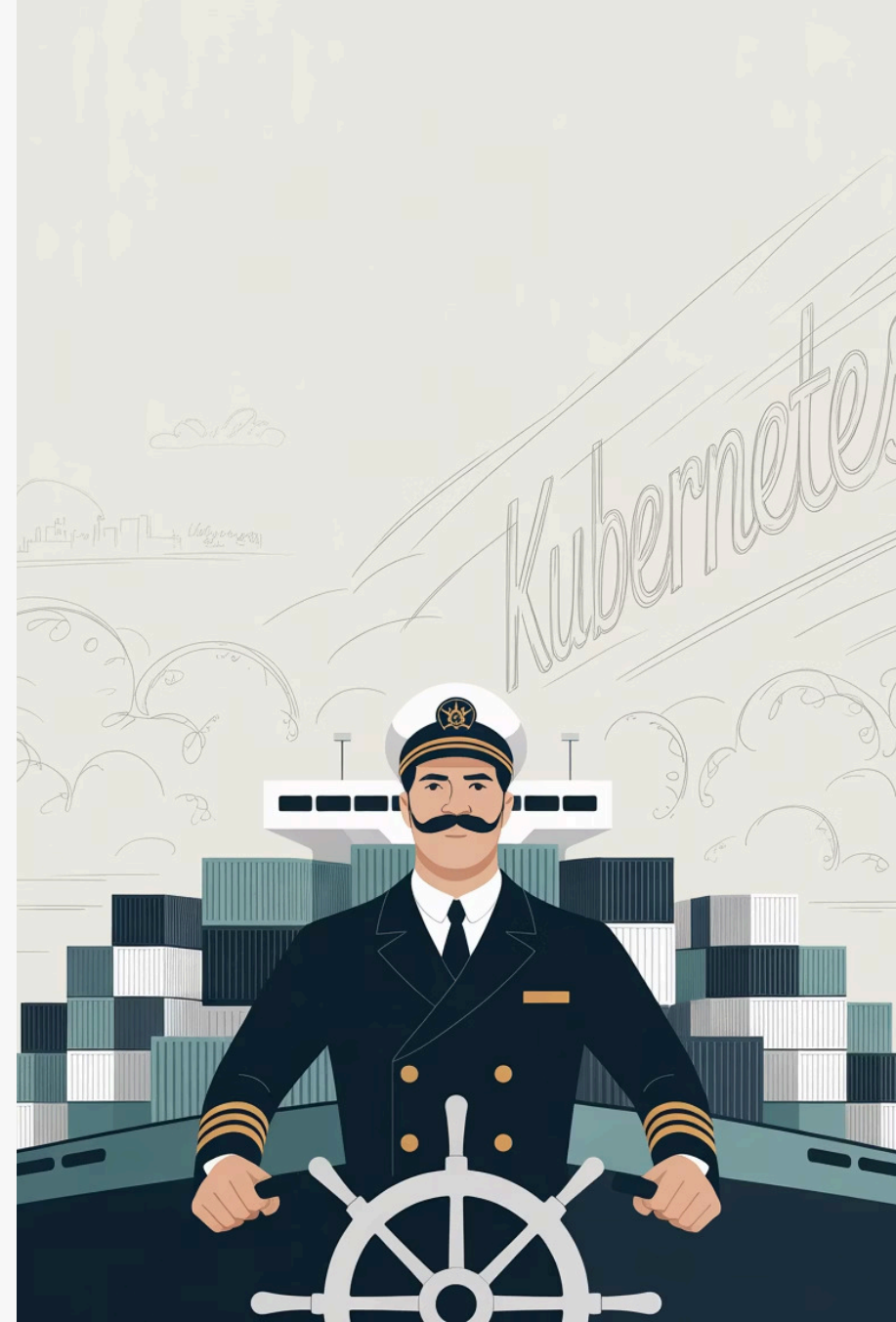
Limitations of Just Docker

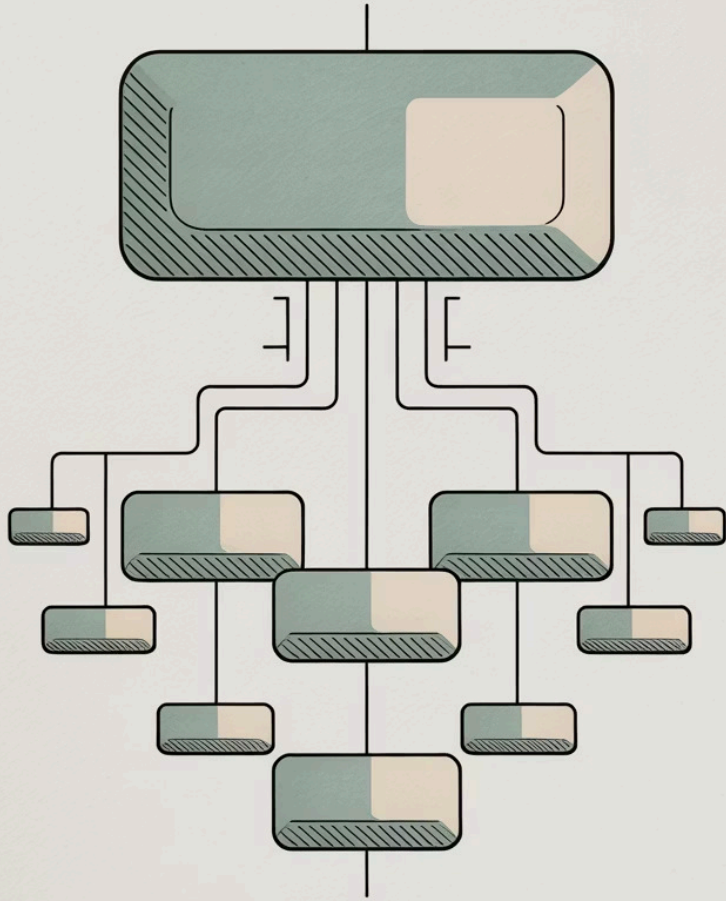
- One app? Easy.
- Many apps? Hard! 😓
- Challenges:
 - Scaling up/down
 - Restarting crashed containers
 - Managing updates

Enter Kubernetes (K8s)

Kubernetes = Container Orchestrator

- Automates deployment & scaling 🤖
- Heals apps when they crash ❤️
- Balances traffic ⚖️





How Kubernetes Works (Simple)

- **Pods** = Groups of containers 🏠
- **Nodes** = Worker machines 💻
- **Cluster** = Group of nodes 🌐
- **Master** = The boss 👑

Docker + Kubernetes Together

- Docker builds containers 
- Kubernetes runs & manages them 
- Like: *Chef cooks the food, Manager serves and organises*  



Real-World Examples

- Google → Scales YouTube videos 📺
- Spotify → Streams music 🎵
- Flipkart/Amazon → Handles festive sales 🛒

Google

 Spotify

flipkart

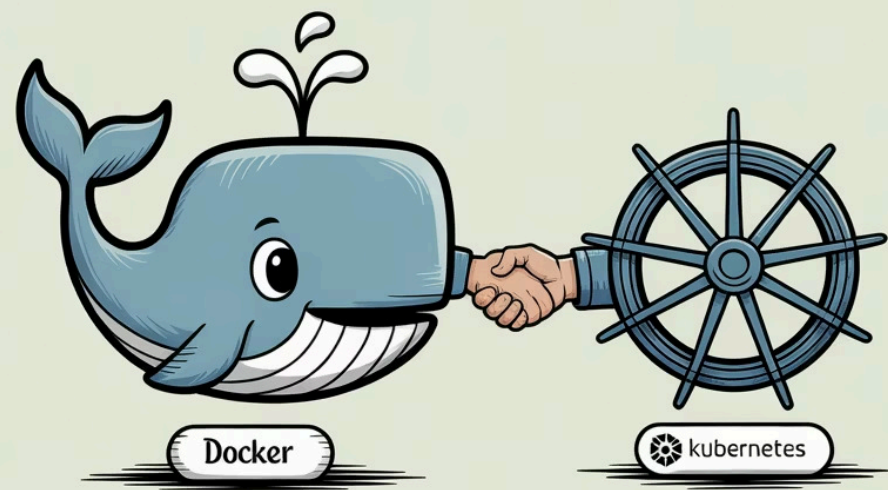
amazon

Charting your Course



🎓 Why Students Should Care

- Cloud jobs demand Kubernetes knowledge 📁
- Open-source, free to learn 🌐
- Perfect skill for DevOps & Cloud careers 🔑



? Q&A + Quick Recap

Recap:

- Docker = Package apps 🐳
- Kubernetes = Manage containers 🌀
- Together → Scalable, reliable apps 🌍



Quick Docker Demo

Goal: Show how an app runs in a container.

Steps:

1. Pick a simple app: e.g., a web page with "Hello, World!" 🌐
2. Build Docker image: `docker build -t hello-app .`
3. Run container: `docker run -p 8080:80 hello-app`
4. Open browser → see your app at localhost:8080 ✅

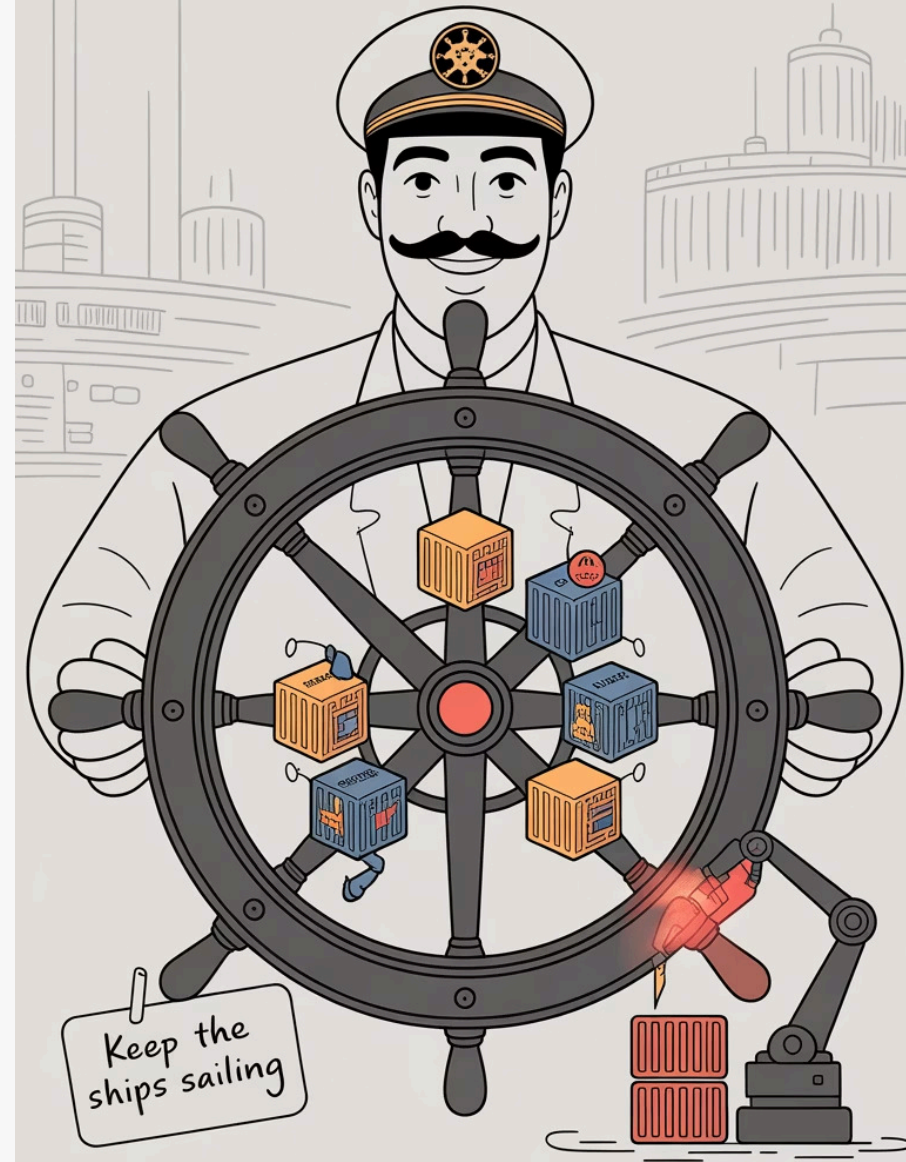


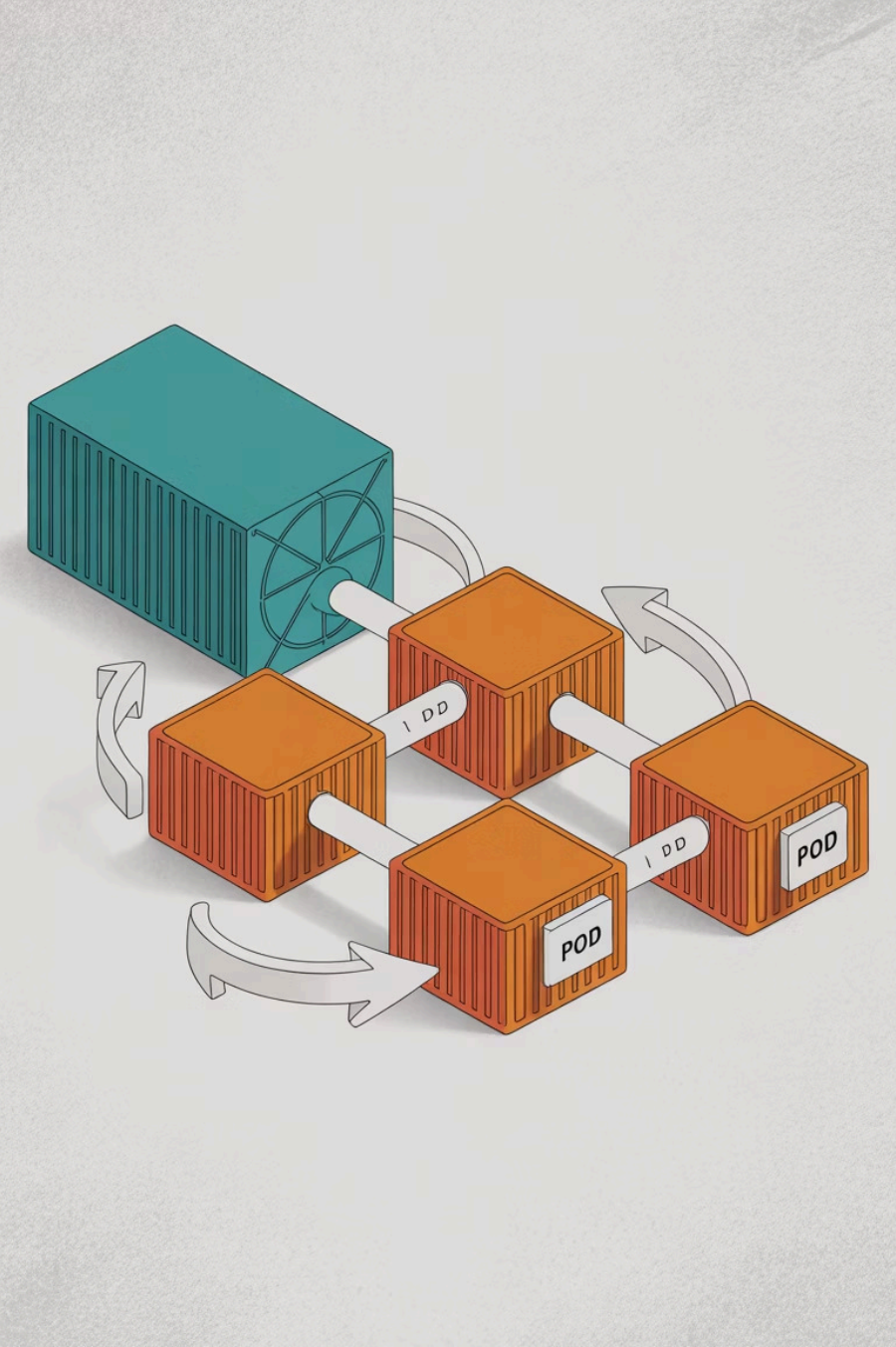
Kubernetes Demo Concept

Goal: Show how Kubernetes manages the same app.

Steps:

1. Create a Kubernetes deployment for your Docker image 
 - `kubectl create deployment hello-app --image=hello-app`
2. Expose it as a service 
 - `kubectl expose deployment hello-app --type=LoadBalancer --port=80`
3. Scale up:
 - `kubectl scale deployment hello-app --replicas=3`
 - Watch 3 copies of your app running automatically
4. Crash one pod → K8s auto-heals it 





Takeaways from Demo

- Docker = Makes your app portable 🐳
- Kubernetes = Makes your app resilient and scalable 🌀
- Together → Apps that "just work" no matter how many users come 🌍